



Estimating genetic variation and heritability of metabolizable energy intake of dairy calves during the rearing period

A. G. R. Phillips,¹ B. O. Mankanjuola,¹ F. Miglior,^{1,2} F. S. Schenkel,¹ M. A. Steele,³ C. F. Baes,^{1*} and R. E. Jahnel¹

¹Centre for Genetic Improvement of Livestock, Department of Animal Biosciences, University of Guelph, Guelph, ON, N1G 2W1, Canada

²Lactanet Canada, Guelph, ON, N1K 1E5, Canada

³Centre for Nutrition Modelling, Department of Animal Biosciences, University of Guelph, Guelph, ON, N1G 2W1, Canada

ABSTRACT

Breeding feed-efficient animals has become a key focus of the dairy industry, yet most research on its genetic potential has focused on adult animals, resulting in a dearth of understanding for feed efficiency in youngstock. Metabolizable energy intake (MEI) of milk replacer and solid feed could provide information on early-life feed efficiency, as it encompasses both forms of the diet fed to young calves. The objective of this study was to estimate the variance components and genetic parameters of MEI in dairy calves over the rearing period (from 5 to 60 d of life). A total of 44,919 daily feed intake records from 913 Holstein calves collected from July 2016 to June 2024 were analyzed. A univariate random regression model was used to estimate genetic parameters of daily MEI. Average heritability and repeatability (SE) for daily MEI were 0.17 (SE = 0.03) and 0.50 (SE = 0.03), respectively, with the highest values for both genetic parameters between d 5 and 18 of life and d 47 to 60 of life. Genetic parameters for daily MEI were lowest during the weaning transition from d 28 to 32 of life. Genetic correlations among daily MEI were highly variable, ranging from −0.46 to 0.99, likely reflecting the physiological changes in digestion as calves transitioned from a liquid to solid diet. We conclude that daily MEI of calves in the rearing period shows potential for genetic selection, with genetic parameters being greatest in the first (wk 1 and 2) and last 2 wk (wk 7 and 8) of life. Further studies with larger datasets and additional herds are needed to confirm these findings.

Key words: genetic parameters, calf, feed intake, feed efficiency

INTRODUCTION

Feeding management during the rearing period is critical for successful calf development. However, feeding protocols are expensive for producers, with about 46% of total costs during the preweaning period associated with feed and feed management (Hawkins et al., 2019). The preweaning period is also regarded as the most expensive phase of rearing replacement heifers due to high costs associated with feeding a milk diet (Hawkins et al., 2019). Restricting milk intake during the rearing period to decrease expenses can adversely affect calf growth (Khan et al., 2011; Soberon et al., 2012), increase susceptibility to disease (Ollivett et al., 2012), and reduce calf welfare (Krachun et al., 2010). Feeding calves at a higher plane of milk and solid feed nutrition is associated with improved preweaning ADG, lactation performance, and lifetime survivability (Gelsinger et al., 2016; Van Amburgh et al., 2019; Leal et al., 2025). Although costly, optimizing calf growth during the rearing period is necessary for optimal calf welfare and future milk production.

A potential strategy to reduce costs without limiting animal development and future performance could be to breed for feed-efficient dairy calves. Feed efficiency (FE) in dairy animals is traditionally defined as an animal's ability to convert feed to milk, with more feed-efficient cows consuming less feed while maintaining milk production or consuming the same amount of feed and producing more milk (Houlahan et al., 2024). Feed efficiency is often measured using residual feed intake (RFI), defined as the residual from a fixed linear regression of DMI on energy sinks, such as milk production and metabolic BW, with positive genetic correlations estimated between heifer and lactating cow RFI (Koch et al., 1963; Kennedy et al., 1993; Gonzalez-Recio et al., 2014). However, genetic selection for FE has primarily focused on adult animals, resulting in limited knowledge regarding its genetic application in youngstock (da Silva et al., 2023). The same understanding of FE in adult animals cannot be directly transferred to youngstock, as

Received August 1, 2025.

Accepted February 24, 2026.

*Corresponding author: cbaes@uoguelph.ca

The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

there are differences in gut function, feeding behavior, and nutrient prioritization (da Silva et al., 2023). For dairy calves, the previous definition of FE can be adapted to describe calves that are more efficient at converting feed into growth, with more feed-efficient calves consuming less feed to achieve the same growth and having the same body maintenance requirements as another calf that consumes more feed.

Accurate definition and measurement of feed intake is essential for determining FE. In adult dairy cattle, DMI is used as the measure of feed intake to derive the RFI phenotype for FE evaluation (Seymour et al., 2019; Negussie et al., 2019; Houlahan et al., 2024). Previous studies have investigated the genetic parameters of DMI and compared the potential of incorporating either DMI or RFI into a selection index to improve FE (Houlahan et al., 2021). However, DMI may not be the most appropriate for FE in dairy calves during the preweaning period. During the first 2 mo of life, calves are fed a combination of liquid and solid diets; using only DMI may fail to account for the energy derived from liquid feed, potentially introducing bias into estimated genetic parameters (Drackley, 2008). Furthermore, calves undergo continuous physiological development within the first 2 mo of life, with growth requirements changing daily and the gastrointestinal tract adapting to digest solid feed (Meale et al., 2017; da Silva et al., 2023). Given these dynamic conditions, ME intake (MEI) from the liquid and solid diet may be a better measure of feed intake to evaluate calf FE, as it reflects true energy utilization from all feed sources (Drackley, 2008).

As a foundation to evaluate calf FE, the dynamics of MEI as a feed intake trait should be investigated. Phenotypic individual variation in milk and calf starter intake has been recorded across various studies, even when animals were housed and raised in similar conditions (de Passillé and Rushen, 2016; Neave et al., 2018). This variation is attributed to differences in behavioral development, season of birth, genetics, health, or dam parity (Heinrichs and Heinrichs, 2011; Miller-Cushon and DeVries, 2015; Neave et al., 2019). According to the Canadian Code of Practice, which recommends calves should consume at least 1.4 kg of starter daily for at least 3 d before weaning, this variation in the start of concentrate consumption could cause differences in weaning age, further affecting growth rates during the rearing period (Quigley et al., 2019; NFACC, 2023). Feed intake variability in the rearing period affects mature weight gain and future milk production, emphasizing the role early-life feed intake plays in long-term productivity for dairy cattle (Soberon et al., 2012; Gelsinger et al., 2016). This variation in feed intake can also delay weaning, resulting in calves remaining on the more expensive milk diet longer and increasing rearing

costs for producers (Hawkins et al., 2019). These findings suggest that MEI needs to be analyzed over the entire preweaning period to better understand the genetic influence behind the variability of calf feed intake.

Gaining understanding of the genetic background of a feed intake trait will provide insight on how to structure a future feed efficiency trait for youngstock. When evaluating longitudinal data, such as daily MEI, repeated measures models, such as random regression models (RRM), are commonly used to estimate variances over time (Jamrozik and Schaeffer, 1997). The RRM have been previously used to estimate genetic parameters of FE traits in adult animals, highlighting the opportunity to apply this methodology to youngstock (Khanal et al., 2022; Houlahan et al., 2024). Therefore, the objective of this study was to estimate the variance components and genetic parameters of MEI in dairy calves over the rearing period using an RRM approach.

MATERIALS AND METHODS

This study was approved by the University of Guelph Animal Care Committee (AUP 4445 and AUP 5290).

Animals

The data used in this study was collected from the Ontario Dairy Research Centre (Elora, ON, Canada) from July 2016 to April 2024. The dataset consisted of 913 Holstein calves with 44,919 records for daily feed intake and 913 records for birth weight. The pedigree included information on 7,878 individuals and 12 generations.

Calves were raised under standard operating procedures at the research station, where they were fed 3 L of colostrum during the first 2 h of life and moved to individual pens, where they remained for the first 4 d of life. Six to 12 h after the first colostrum feeding, animals were offered another 3 L of colostrum. Calves were fed their dam's colostrum, unless the quality was too low, in which case frozen colostrum ($\geq 22\%$ Brix) was fed. Colostrum was offered by bottle but fed via an esophageal tube feeder if the calf did not voluntarily suckle the entire 3 L from the bottle. On d 1 of life, calves were offered 3 feedings of their dam's transition milk in 2-L quantities by bottle. From d 3 of life onward, calves were fed milk replacer at a concentration of 150 g/L (26% CP, 18% crude fat; Excel Pro-Gro Milk Replacer Powder, Grober Nutrition, Cambridge, ON, Canada).

On d 5 of life, calves were moved to grouped housing in 1 of 2 calf nursery rooms at the research station (**R1** and **R2**, respectively), where each room had an automated milk and calf starter feeder (**MR** and **CS**, respectively; Forster Technik). Calves were housed to a maximum of 15 calves per room, with animals leaving the

nursery after they reached ~2 mo of age. Calves were fitted with radio frequency identification (RFID) tags that allowed or denied access to feeders, and quantity fed per visit was recorded. The automatic milk feeder was programmed to include a milk entitlement for each calf, with the consumption entitlement changing every day based on the date the calf first entered the system, as well as carryover based on the previous day. Calves were fed the previously mentioned milk replacer at 150 g/L on the automated feeder until weaning started on d 28 of life for each calf, regardless of individual solid feed intake. During the weaning process, calves were restricted to a maximum intake of 12 L/d and gradually reduced by 0.5 L/d until weaning ended on d 57 of life.

The maximum meal size for milk replacer from the automatic feeder was 2.5 L from d 5 to 21 of life, then 3 L from d 22 to 63 of life. On d 28 of life, if a calf drank their maximum meal size at one visit, they would have to wait at least 2 h before they were allowed to drink again. Once weaning began on d 28 of life, calves were then limited to the total amount of milk per day they were allowed to consume based on the step-down weaning protocol. In the group-housed pens, calves also had free access to an automated concentrate feeder providing a 22% calf starter (22.1% CP, 27.63% starch, 14.27% NDF pellet; Floradale Feed Mill Limited, Floradale, ON, Canada). Calf starter was dispensed in 30-g increments to a maximum of 6 kg/d. The calf starter was offered to each calf when they visited the feeder, however, there was no scale to weigh refusals and quantify if any feed had been left behind by the visiting calf. Therefore, the solid feed intake records we use in this analysis are on an as-offered basis.

The concentration of the milk replacer provided to calves was 150 g/L of water, with 4.55 Mcal of ME per kilogram of powder. The pelleted calf starter contained 88% DM and 2.51 Mcal/kg ME on a DM basis. Values for the ME component of diets were provided by the feed suppliers. Feed provided was assumed to be consistent over the study period. For the composition of the diets fed, see Supplemental Tables S1 and S2 (see Notes). Of all the calves in the population, 120 were part of a calf starter trial where animals were assigned to 1 of 2 milk treatments (low or high level using the automatic milk-feeding system) and 1 of 2 feed treatments (milk byproduct starter pellet with 3.4 Mcal/kg or grain-based feed with 3.1 Mcal/kg; Parsons et al., 2022). Calves in the previously mentioned trial were fed in individual pens using a calf rail rather than the standard automatic milk-feeding system (Parsons et al., 2022). The effect of the feed provided to the animals in this trial study was accounted for in the model.

At each visit, the automatic feeders recorded the calf's RFID, the date and time they visited the feeder, and

the amount of feed that was consumed (milk replacer) or offered (calf starter) during that visit. Feed intake records from the automatic feeders were cumulative for each day, meaning that each calf's last daily record from the feeder was the total amount of milk replacer or starter the calf had consumed that day. In total, there were 24,973 daily intake records from R1-MR, 24,003 daily intake records from R1-CS, 30,878 daily intake records from R2-MR, and 30,119 daily intake records from R2-CS. Combined between the 2 nurseries, there were 54,122 initial calf starter intake records and 55,851 initial milk replacer intake records. Daily intake records from less than 5 d of life and greater than 60 d of life were removed, as calves were assumed to have their first intake record on d 5 and were assumed to have moved out of the nursery on d 60 of life at 2 mo of age (3,000 milk intake and 2,694 calf starter intake records, respectively).

Intake records were converted to the powder component of milk replacer and the DMI of the calf starter. The concentration of the milk replacer was used to convert the milk replacer into its powder component. The DMI of the calf starter was calculated using the DM component of the diet provided by the feed supplier. The milk replacer powder and calf starter DM components were then transformed into MEI using the ME component of the respective diet. Total daily MEI was calculated from the sum of the milk replacer MEI and concentrate MEI for each day. Duplicate records were removed based on the calf's RFID, date of birth, and age, resulting in a combined total of 51,377 records. Daily MEI records that were less than or equal to zero and records that were greater than 60 d of age were removed, resulting in 44,919 daily MEI records for the final analysis. Total test days in this analysis ranged from d 5 to 60 of life, with each calf having up to 56 daily records. For the distribution of daily MEI records across the first 2 mo of life, see Supplemental Figure S1 (see Notes).

Birth weight records were collected manually for each calf, and metabolic BW (MBW) at birth was calculated as (birth weight)^{0.75}. The MBW at birth was used rather than just birth weight to capture the metabolic requirements of the calf, similar to the use of MBW to evaluate DMI and FE in adult dairy cattle (Houlahan et al., 2021). There were 23 calves who were weighed after their actual birthdate, with manual weight records taken between 1 and 3 d later.

Diarrhea was diagnosed by farm staff based on visual diagnosis and rectal temperature. If a calf met the visual diagnosis criteria, they were treated with electrolytes twice per day until diarrhea improved. If a calf also had a high temperature, they received a subcutaneous injection of 0.5 mg/kg meloxicam.

Variance Component Estimation

Additive genetic, permanent environmental, and residual variances were estimated using a univariate random regression animal model in ASReml 4.2 (VSNi; Gilmour et al., 2015). Data were analyzed using R software (R Core Team, 2022). The model used is described as follows:

$$y_{iwgjn} = YS_i + age_w + t_g + s_j + \beta_2 MBW_c + \sum_{k=0}^3 \phi_{t,k} a_{c,k} + \sum_{k=0}^4 \phi_{t,k} pe_{c,k} + e_{iwgjn},$$

where y_{iwgjn} denotes the n th test day record of calf c ; YS_i is the fixed effect of the i th year-season; age_w is the categorical effect of age (week) w ; t_g is the fixed effect of trial g (5 levels); s_j is the fixed effect of daily diarrhea incidence j ; β_2 is the fixed linear effect of the MBW at birth of calf c ; $\phi_{t,k}$ is the (normalized) Legendre polynomial coefficients ($k = 0, \dots, 4$), evaluated at age t , that is,

$$\begin{aligned} \phi_{t,0}(z) &= \sqrt{\frac{1}{2}}, \phi_{t,1}(z) = \sqrt{\frac{3}{2}}z, \phi_{t,2}(z) = \sqrt{\frac{5}{2}}\frac{1}{2}(3z^2 - 1), \\ \phi_{t,3}(z) &= \sqrt{\frac{7}{2}}\frac{1}{2}(5z^3 - 3z), \phi_{t,4}(z) = \sqrt{\frac{9}{2}}\frac{1}{8}(35z^4 - 30z^2 + 3), \end{aligned}$$

with $z = -1 + 2\left(\frac{t - t_{min}}{t_{max} - t_{min}}\right)$ (Kirkpatrick et al., 1990);

$a_{c,k}$ and $pe_{c,k}$ are the random regression coefficients for calf c accounting for additive genetic effects and permanent environmental effects, respectively; and e_{iwgjn} is the residual. All fixed effects in the model were significant at a 5% level. Residuals of the model were fit with heterogeneous variance in 8 weekly time periods to cover the 2-mo period that daily MEI was recorded. The weekly time periods were: d 5 to 11, 12 to 18, 19 to 25, 26 to 32, 33 to 39, 40 to 46, 47 to 53, and 54 to 60. Model fit with different degrees of Legendre polynomials was tested, and the best model was selected based on Akaike information criterion (AIC) and on producing biologically and genetically sound parameters with no signs of over-parameterization (Supplemental Figure S2, see Notes).

It was assumed that $var(a) = G \otimes A$, $var(pe) = C \otimes I_c$, and $var(e) = I_e \sigma_e^2$, G and C are the (co)variance components for genetic and permanent environmental Legendre polynomial coefficients, respectively; $\otimes$ denotes the

Kronecker product; I_c is an identity matrix of order equal to the number of calves with records; and I_e is an identity matrix of order equal to the sample size. If $\phi_t = (\phi_{t,0}, \phi_{t,1}, \phi_{t,2}, \phi_{t,3})$ is the vector of Legendre polynomials, the additive genetic variance was obtained at age t was $\sigma_a^2 = \phi_t G \phi_t'$ and the permanent environmental variance was $\sigma_{pe}^2 = \phi_t C \phi_t'$. Thus, the estimated narrow-sense heritability (h_t^2) and repeatability (r_t^2) at age t were

$$\widehat{h}_t^2 = \frac{\phi_t G \phi_t'}{\phi_t (G + C) \phi_t' + \sigma_e^2} \quad \text{and} \quad \widehat{r}_t^2 = \frac{\phi_t (G + C) \phi_t'}{\phi_t (G + C) \phi_t' + \sigma_e^2}.$$

The SE of genetic parameters was approximated at age t by first-order Taylor expansion, adapted from Jahnke et al. (2023) and Fisher et al. (2004). The SE for heritability estimates was calculated as

$$var(\widehat{h}_t^2) \approx \nabla f_t(\sigma)' \Sigma \nabla f_t(\sigma) \quad \text{and} \quad SE(\widehat{h}_t^2) = \sqrt{var(\widehat{h}_t^2)},$$

where $var(\widehat{h}_t^2)$ is the variance of the narrow-sense heritability h_t^2 , $\nabla f_t(\sigma)'$ is the gradient of the function f_t with respect to σ , and $\Sigma \nabla f_t(\sigma)$ is the delta method approximation for f_t with respect to σ . The SE for repeatability estimates was calculated as

$$var(\widehat{r}_t^2) \approx \nabla g_t(\sigma)' \Sigma \nabla g_t(\sigma) \quad \text{and} \quad SE(\widehat{r}_t^2) = \sqrt{var(\widehat{r}_t^2)},$$

where $var(\widehat{r}_t^2)$ is the variance of the narrow-sense repeatability r_t^2 , $\nabla g_t(\sigma)'$ is the gradient of the function g_t with respect to σ , and $\Sigma \nabla g_t(\sigma)$ is the delta method approximation for g_t with respect to σ . The genetic correlation between MEI at different ages t_1 and t_2 was

$$\widehat{r}_{g,t} = \frac{\phi_{t1} G \phi_{t2}'}{\sqrt{(\phi_{t1} G \phi_{t1}')(\phi_{t2} G \phi_{t2}')}}.$$

To investigate the genetic trend in daily MEI, EBV of phenotyped animals were obtained as

$$EBV_t = \frac{1}{56} \sum_{t=1}^3 \sum_{k=0}^3 \phi_{w,k} u_k,$$

and the EBVs were then regressed onto the birth year to determine the annual genetic change.

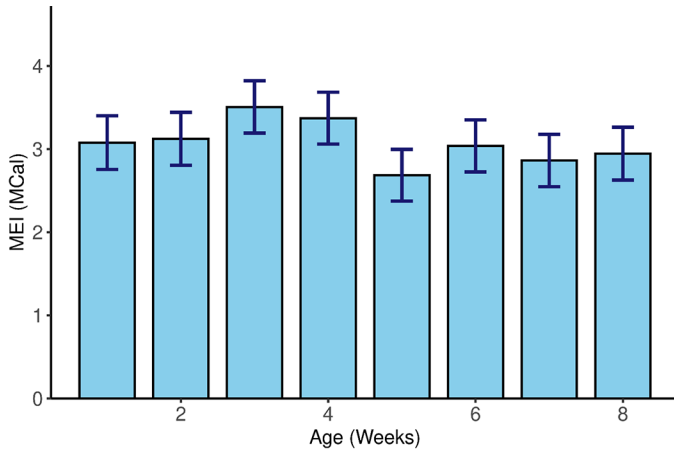


Figure 1. Metabolizable energy intake (in MCal) over time (age in weeks) for preweaning dairy calves. Error bars are SE.

RESULTS

Descriptive Statistics

Descriptive statistics of the data used in this study are presented in Table 1. On average, an individual calf had 49 daily MEI records. Average daily MEI was 3.08 MCal (SD = 0.27), with the lowest estimates for MEI found at 5 wk of age (Figure 1). The MEI increased steadily from 3.08 MCal (SD = 0.27) at 1 wk of age before peaking at 3.51 MCal (SD = 0.27) at 3 wk of age. Average birth weight and MBW at birth were 40.93 kg (SD = 4.58) and 16.16 kg^{0.75} (SD = 1.36), respectively. A total of 205 calves had diarrhea across the 2-mo study period, and 36 of those calves had more than one diarrhea incidence. Diarrhea incidence was highest in the first 4 wk of study, with the highest incidence rate on d 11 of life.

Variance Components of Daily MEI

Average daily genetic, permanent environmental, and residual variances were 0.47 (SE = 0.10), 0.92 (SE = 0.07), and 1.25 (SE = 0.03), respectively. Genetic parameters of daily ME intake are presented in Figure 2. The permanent environmental and residual variance were greater than the additive genetic variance across the rearing period, except for the first and eighth wk of life (Figure 2A). Permanent environmental variance increased between d 10 and d 15 of life, before gradually decreasing until d 42 of life. The permanent environmental and additive genetic variances were the greatest at the beginning and end of the rearing period. The permanent environmental and additive genetic variance were lowest from d 31 to 41 of life. The heterogeneous residual variance increased within the first 3 wk of life, peaked in wk 4, and then decreased by wk 5 of life. Estimates

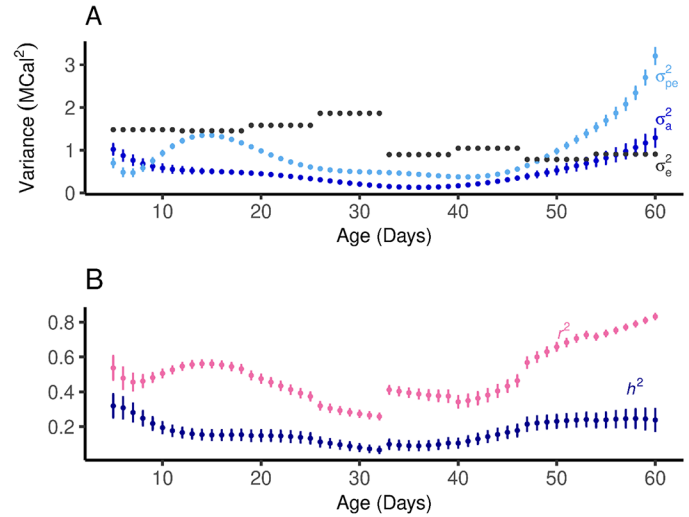


Figure 2. Genetic parameters of daily ME intake (MEI). (A) Genetic (σ_a^2), permanent environmental (σ_{pe}^2), and residual (σ_e^2) variances and their SE as error bars over the first 2 mo of life. (B) Heritability and repeatability estimates and their SE as error bars for daily MEI over the first 2 mo of life.

of residual variance in the first month were greater than estimates in the second month of the rearing period.

Heritability and Repeatability of Daily MEI

The average heritability for daily MEI over the first 2 mo of life was 0.17 (SE = 0.03), with an average estimated genetic SD of 0.69 MCal. Heritability estimates for daily MEI fluctuated across the 2-mo time period, ranging from 0.07 (SE = 0.02) to 0.32 (SE = 0.07; Figure 2B). Heritability estimates for daily MEI were highest on d 5 of life and gradually decreased, reaching a minimum on d 32 of life. Around d 40 of life, heritability estimates began to increase until d 60. The heritability curve followed a similar trajectory to the additive genetic variance. The average repeatability for daily MEI over the first 2 mo of life was 0.50 (SE = 0.03). Repeatability estimates for daily MEI ranged from 0.26 (SE = 0.02) to 0.83 (SE = 0.01; Figure 2B). Estimates were highest at the beginning and end of the rearing period. Repeatability estimates were lowest between d 28 and 32 of life. For the average heritability and repeatability estimates for each week of MEI, see Supplemental Table S3 (see Notes).

Genetic and Phenotypic Correlations of Daily MEI

Genetic correlations between daily MEI estimates ranged from -0.46 to 0.99. For the average genetic correlations between each week of MEI, see Supplemental Table S4 (see Notes). Strong positive genetic correlations were estimated between consecutive daily MEI

Table 1. Descriptive statistics for birth weight, metabolic BW at birth, and daily ME intake (Mcal) for the calf population investigated

Item	Records (n)	Mean	SD	CV	Minimum	Maximum
Birth weight (kg)	913	40.93	4.58	11.19	25.00	59.90
Metabolic BW at birth (kg ^{0.75})	913	16.16	1.35	8.43	11.18	21.53
Daily MEI (Mcal)	44,919	3.08	0.27	8.62	2.69	3.51

records across the rearing period (Figure 3A). Genetic correlations began to decrease when time between daily measurements exceeded 7 d. Daily MEI records from wk 2 and 3 of life showed moderate negative genetic correlations with records from wk 6 to 8 of life. Phenotypic correlations estimated between daily MEI records also varied across the rearing period, with correlations ranging from -0.30 to 0.81 (Figure 3B). Similar to the genetic correlations, strong positive phenotypic correlations were observed between consecutive daily MEI estimates and began to weaken past weekly time frames, with low negative phenotypic correlations observed between early and late daily MEI records.

Phenotypic and Genetic Trends

Over the 8-yr period covered in this study, yearly phenotypes for daily MEI showed little variation (Figure 4A). Genetic trends based on the EBVs of phenotyped animals were also relatively stable (Figure 4B). The genetic trend did become more positive from 2016 to 2017 and then showed minimal change from 2017 onward to 2024.

DISCUSSION

In this study, we estimated the genetic parameters of MEI using an RRM approach to better understand how genetics may influence calf feed intake during the rearing period. The findings from this study provide the genetic background for a measure of feed intake that could be used to derive calf FE. Daily MEI was lowest at the beginning of the rearing period and increased steadily, peaking ~ 4 wk of age. During this time, calves were fed an ad libitum milk replacer diet while consuming minimal solid feed, as solid feed intake is typically limited due to rumen immaturity in early life (Drackley, 2008; Hammon et al., 2020). Young dairy calves are highly motivated to drink large amounts of milk before the intake of solid feed before stabilizing at ~ 4 wk of age (de Passillé and Rushen, 2016; Meale et al., 2017). Notably, the amount of milk calves will drink when allowed ad libitum access varies between calves (Hammon et al., 2002; Soberon et al., 2012). Calves that drink large amounts of milk within the first days of life achieve greater growth rates before and after weaning (Borderas et al., 2009; Rosenberger et al., 2017). This motivation to drink large volumes of

milk early in life reflects the higher MEI values estimated within the first 4 wk of life in this study. Furthermore, the milk replacer diet fed to the calves in this study had double the ME content available compared with the solid diet they were weaned onto. On d 28 of life, milk replacer intake was restricted as the weaning process began. The decline in MEI after the fourth week of age likely reflects this change in diet, as well as individual variation of solid feed intake within the calf population.

Automatic feeders have been shown to achieve similar or improved starter intake and weights after weaning for dairy calves compared with conventional methods (Benetton et al., 2019). Nonetheless, age of weaning based on starter intake still varies with either automatic feeders or conventional methods (Roth et al., 2009). Not every calf is successfully weaned using an automatic system; Welk et al. (2022) found that even with individual intake-based weaning using automatic feeders, 10 out of 108 calves failed to meet preweaning starter intake targets. Reduced starter intake and final BW of a subset of calves weaned on automatic feeders was noted in other studies, demonstrating that there are still unknown variables affecting pre- and postweaning solid intake in calf rearing (de Passillé and Rushen, 2012; Benetton et al., 2019). Due to the individual variability of weaning success, it may be necessary to collect more MEI records during the weaning transition to improve the accuracy of genetic estimates.

In total, 22.5% of calves had diarrhea at some point during the rearing period, with 17.5% of those calves experiencing more than one episode. The incidence rate in this study was slightly higher than the incidence rate on Ontario dairy farms in 2020; this could be possibly related to better disease incidence recording at a research center compared with a commercial setting (Hyland et al., 2022; van Staaveren et al., 2023). Digestive disorders, such as diarrhea, commonly occur within the first 2 wk of life. Although diarrhea is associated with calf susceptibility to different bacteria and viruses, nutritional diarrhea can also occur when calves drink excess quantities of milk (Bazeley, 2003). Regardless of the cause, diarrhea can depress feed intake, which negatively affects early-life weight gain and mature milk production (Abuelo et al., 2021). Calves with diarrhea may have had lower phenotypic MEI records, which could impact genetic parameter estimates of MEI earlier in the rearing

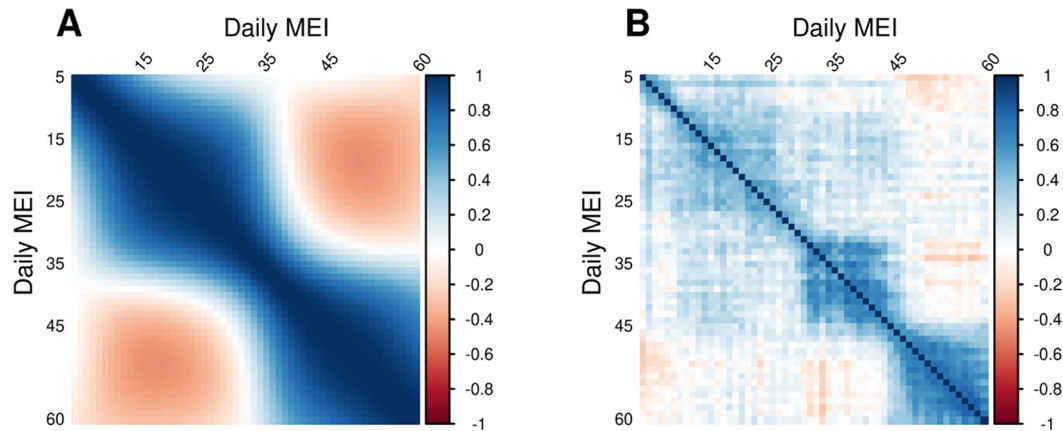


Figure 3. Heatmap of (A) genetic correlations between daily ME intake (MEI) across the first 2 mo of life and (B) phenotypic correlations between daily MEI across the first 2 mo of life. Axis labels run from d 5 to 60 of life (each pixel represents 1 d).

period, even if the occurrence of diarrhea was included as an effect in the evaluation model.

Genetic Parameters of Daily MEI

Both permanent environment and residual variances were higher than the additive genetic variance of MEI across most of the rearing period. However, the additive genetic and permanent environmental variances were high at the beginning and end of the rearing period. It is likely that this is a result of using Legendre polynomials, which tend to be inflated at the extremes of the estimated trajectory (Oliveira et al., 2019). In this study, third-degree and fourth-degree Legendre polynomials were used to estimate the additive genetic and permanent environmental effects for MEI, respectively. Third-degree Legendre polynomials appear appropriate to estimate the additive genetic (co)variances for FE traits in adult Holstein cows (Togashi et al., 2008; Houlahan et al., 2024). Recent work by Graham et al. (2024) found that using second-degree Legendre polynomials was sufficient to estimate the additive genetic (co)variances of calf milk intake traits. The model in this study that used third-degree and fourth-degree Legendre polynomials showed much lower AIC than alternative lower polynomial degrees. The higher polynomial degree likely captures the more complex pattern of genetic and especially permanent environmental variance in calf MEI. Future studies should further evaluate alternative polynomial degrees when modeling feed intake traits in calves.

Heterogeneous residual variances were used in this study to analyze MEI over the rearing period. Previous studies have emphasized that improved goodness of fit can be achieved using heterogeneous residual variances across different time periods (Pereira et al., 2013; Flores and van der Werf, 2015). Recent work by Graham et

al. (2024) also found that random regression models with heterogeneous residuals had superior performance compared with alternative random regression and repeatability models to capture the genetic variance of milk consumption traits in calves. The decrease in residual variances after the first month of the rearing period could be associated with the start of milk replacer restrictions on d 28 of life. With this restriction on milk intake, the variability within individual MEI records likely decreased as calves consumed more uniform allotments of milk replacer. Variability in preweaning milk intake has been observed in other studies and has been associated with calves having differing levels of motivation to access milk (Rosenberger et al., 2017; Neave et al., 2019). Neave et al. (2019) also noted variability in motivation among individual calves to access milk, with unrewarded visits to an automatic feeder ranging from 211 to 1,160 within 105 d of study. Heterogeneous residuals likely also reflect the different diets given to calves throughout their first developing stage in the rearing period.

It is important to consider both the RRM approach and sample size used to derive genetic estimates for MEI. Random regression models use longitudinal data, which allows estimates to change over the studied time period (Schaeffer, 2004). Furthermore, feed intake during the preweaning phase of the rearing period is low and variable among and within individual calves, which could make estimating differences in feed intake difficult (da Silva et al., 2023). This study used RRM but only had a small sample size for genetic parameter estimation, as well as records from only one herd. Future research should be conducted with multiple herds and a larger calf population to confirm these findings. As of 2017, 16% of Canadian commercial dairy herds had adopted the use of automatic milk feeders for calf rearing (Medrano-Galarza et al., 2017). This presents an opportunity to collect

further MEI phenotypes across the country to improve upon initial estimates made in this study.

The low to moderate heritability estimates for daily MEI of calves show similar patterns to heritabilities estimated for feed intake in adult cows. Karacaören et al. (2006) found that the heritability for DMI in first-lactation cows ranged from 0.12 to 0.34, with an average heritability of 0.23. Heritability estimates for DMI in first-lactation Holstein cows ranged from 0.15 to 0.29 across the 305-d lactation period (Houlahan et al., 2024). In this study, heritability estimates of MEI increased toward the end of the study period, which could be related to calves having a more consistent feed intake. Before calves begin the weaning transition, they may only consume small amounts of solid feed as they still prefer the liquid diet (Miller-Cushon et al., 2013b). As calves become more familiar with concentrate feed during the weaning period, the temporary environmental influence may begin to decrease (Khan et al., 2016). Extending the period of study past 60 d of age may provide more insight into this increase in heritability estimates as calves consume solid feed more consistently. The variability of heritability estimates may also reflect the number of records available for specific time periods, as a smaller amount of MEI records were available in the first few weeks of life compared with later in the rearing period (Supplemental Figure S1). In particular, the number of daily records was the highest from d 45 to 60, which could be involved with the increase in heritability estimates toward the end of the rearing period. Fewer daily MEI records early in the rearing period could be associated with calves learning how to independently access automatic feeders. Medrano-Galarza et al. (2018) found that calves moved into group housing on d 5 of life took an average of 31.4 h to successfully learn how to use the feeder. Jensen (2007) found that 6-d-old calves on a restricted milk allowance required 2 d on average (interquartile range: 1–4 d) to learn how to drink from the automatic feeder. This latency to learn how to access the automatic feeders independently likely contributes to the smaller number of records earlier in the rearing period.

Repeatability estimates for daily MEI were dynamic across the rearing period and were lowest during the weaning transition period. The lower repeatability during this time may reflect each individual calf's ability to smoothly transition from the liquid to solid diet. Weaning can be physiologically stressful for calves, with the impact of weaning stress varying among individual animals (Khan et al., 2016). Due to the individual variability of weaning success, it may be necessary to collect more MEI records during the weaning transition to improve the accuracy of genetic estimates. Variation in repeatability estimates could also stem from differences in where the feed is being digested in the calf's gastrointestinal

tract. As the calf is eating during this developmental period, the liquid diet is directed into the abomasum through the pharyngeal reflex, and the concentrate feed is directed to the rumen (Meale et al., 2017). However, as calves prefer the liquid milk diet, especially early in life, overconsumption of milk can have a negative impact on rumen maturation. Calves that are solely fed a liquid diet have stunted rumen musculature and weight, which adversely affects their ability to properly digest solid feed as they mature (Diao et al., 2019). Sometimes, overfeeding milk can also cause the liquid diet to spill into the rumen, which can ferment and alter the rumen microflora (Ellingsen et al., 2016). As the overconsumption of the liquid diet can affect solid feed intake and maturation of the rumen later on, this may affect phenotypic MEI and genetic estimates later in the rearing period. Repeatability estimates of MEI may potentially improve if the MEI of the liquid and solid diets were viewed as separate traits across the rearing period.

Genetic and Phenotypic Correlations of Daily MEI

As a feed intake trait, MEI is likely controlled by biological processes that affect the observed phenotype across the rearing period. Phenotypic variations of feed intake have been observed within individual dairy cows, potentially suggesting that feed intake is a genetically different trait across the lactation period (Strucken et al., 2015; Li et al., 2018). To identify whether the same phenomenon occurs with MEI during the rearing period, the potential changes in genetic and phenotypic correlations across the rearing period were investigated.

Strong positive genetic correlations between daily MEI recorded within 1-wk periods were observed in this study. The strong positive genetic correlation between daily MEI estimates in 1-wk periods suggests that weekly MEI could be used instead of daily measurements to evaluate MEI. As feed intake across the rearing period is quite low compared with adult animals, it is likely that differences between daily intakes are negligible compared with time periods that are farther apart (da Silva et al., 2023).

Moderate negative genetic correlations were observed between daily MEI estimates around early and late age in the rearing period. These correlations suggest that MEI is a genetically different trait over time, supported by the rapid transformation in digestive physiology that dairy calves experience during the first 2 mo of life. Within the first month of life, the abomasum is responsible for digesting the liquid feed, as the rumen is premature and has limited function (Meale et al., 2017). Before eating solid feed, the calf gastrointestinal tract (GIT) functions as that of a monogastric animal rather than a ruminant (Khan et al., 2016; Ghaffari et al., 2025). As calves begin to eat solid feed throughout the rearing period,

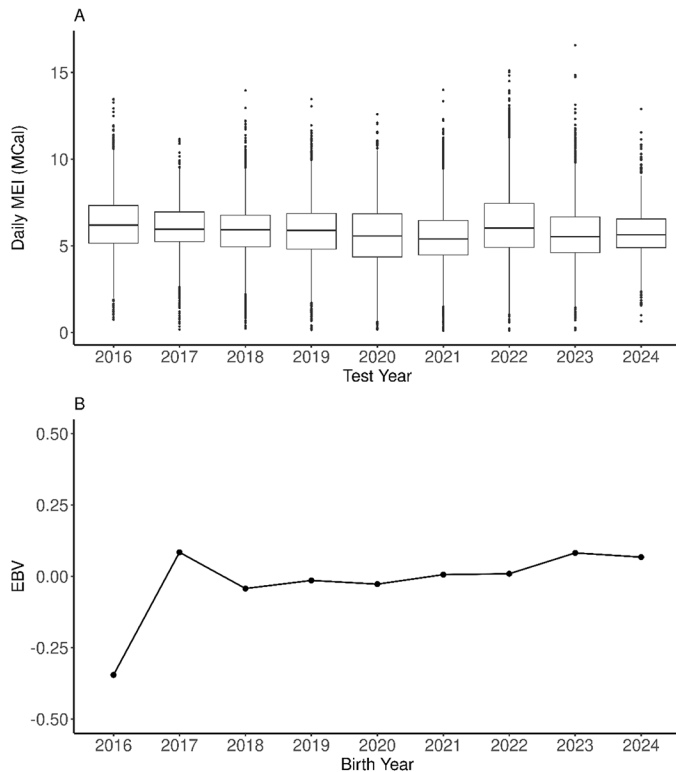


Figure 4. (A) Phenotypic distribution of daily ME intake (MEI) by test year with lower and upper edges representing the 25th and 75th percentiles, midline indicating median of data, whiskers extending to $1.5 \times$ interquartile range (IQR), and dots representing outliers. (B) Genetic trend for 60-d EBVs of phenotyped animals for daily MEI by year of birth.

the rumen begins to mature, and its function improves. From birth to 8 wk of age, the reticulorumen increases from 38% to 61% of total stomach mass, surpassing the size of the abomasum (Diao et al., 2019). Along with increasing in mass, rumen physiology rapidly adapts to properly digest solid feed. The solid feed consumed by the calf contributes to a decrease in rumen pH and an increase in VFA concentration (Khan et al., 2016). Volatile fatty acid concentration, primarily butyrate, acetate, and propionate, contributes to the length and proliferation of rumen papillae, which is essential to increase surface area for nutrient absorption (Yonekura et al., 2002; Baldwin et al., 2004). It has been hypothesized that calves differ genetically in rumen development, with some calves being more fit to digest feed either in the preweaning or weaning stages of the rearing period (Roth et al., 2009). The moderate negative correlations could also imply that the MEI of milk replacer and concentrate should be treated as separate traits, given the difference in GIT physiology required to successfully digest either diet. Another possibility could be to group MEI into intake periods covering the first and second

month of life, in which calves are primarily consuming either liquid or solid feed.

Phenotypic correlations exhibited similar relationships, as seen with estimated genetic correlations. Moderate negative phenotypic correlations were estimated between early and later MEI records, starting around d 45 of life. This negative relationship could be the result of high milk intake suppressing solid feed intake later in the rearing period. Calves start consuming adequate amounts of solid feed starting at 2 wk of age, with intake rapidly increasing when milk allowance is reduced (Khan et al., 2011). Inverse relationships between milk and solid feed intake have been observed across studies (Khan et al., 2007; Sweeney et al., 2010; Silva et al., 2019). Calves that consume greater volumes of milk are likely less hungry, as they are satiated from the liquid diet (Khan et al., 2011). Previous studies have found that with every increase of 100 g in liquid feed DMI, a reduction of 13 to 93 g of solid feed DMI is expected (Silva et al., 2019). This suppressive relationship has been shown to delay rumen development, which impacts digestion later in life (Sweeney et al., 2010; Miller-Cushon et al., 2013a; Benetton et al., 2019).

Genetic Trends

The results of this study revealed no consistent trend in EBVs for daily MEI among phenotyped animals. This lack of directional change may reflect the current emphasis in FE research and selection on heifers and adult cows, rather than preweaning calves. Most FE studies have focused on trait expression in later stages of development and on genetic correlations between heifer and lactating cow feed intake (Pryce et al., 2015; Khan et al., 2024). However, early life traits have been shown to significantly influence long-term productivity and performance within the milking herd (Van De Stroet et al., 2016). Enhancing our understanding of feed intake dynamics in the early stages of life could enable selection for more efficient youngstock and potentially improve lifetime efficiency in dairy cattle.

Limitations

In this study, we used pedigree information to estimate the genetic parameters of MEI across the rearing period. Future studies should incorporate genomic information to improve the accuracy of genetic parameter estimates and EBVs (Hayes et al., 2013). Including genotypic information would enable estimation of the observed relationship between animals. Graham et al. (2024) used genomic information to estimate genetic parameters for feeding behavior traits in dairy calves and reported

a higher average heritability (0.46) for daily milk consumption compared with the heritability found in this study, as well as a moderate reliability of 0.5. Milk consumption in the Graham et al. (2024) study was recorded from d 1 to d 60 of life, and birth weight (in kilograms) rather than MBW at birth was fit as fixed effect in the RRM. The findings from Graham et al. (2024) highlight the impact genomic information could have on breeding for more robust and efficient calves.

An RRM approach was used to model the trajectory of MEI over the rearing period in this study, whereas an alternative approach would be to use a multiple-trait model for MEI in different rearing periods. Previous studies have compared the use of RRM and multiple-trait models for growth and weight traits in beef cattle and sheep, with differing results in the improvement of EBV accuracy and genetic parameter estimates (Wolc et al., 2011; Khorshidi et al., 2020; Paneru et al., 2024).

The use of automated calf feeders increased the amount of individual phenotype records that we could use to estimate genetic parameters of daily MEI. Previous research has used automatic milk feeders for phenotype collection to derive milk intake and disease resilience traits in dairy calves (Graham et al., 2024). Although the accuracy and validity of automatic milk feeders has been discussed in literature (Costa et al., 2021), there are no available validation studies focused solely on solid feeders. The use of the automated calf starter feeders reduced labor requirements for data collection but did introduce error into this study. The pelleted feed was dispensed in 30-g increments, however, there was no information on whether the calf consumed all feed dispensed or whether the calf ate leftover feed after a previous calf left the feeder. This could also explain the lower heritability and repeatability in the transition period, from liquid to solid diet, observed in this study.

Within the 2-mo period of study, multiple calves had missing daily MEI records. Missing daily feed intake records could indicate that the animal did not visit the automatic feeder on that day and has been attributed to sickness behavior, heat stress, or satiety from earlier meals in previous studies (Conboy et al., 2022; Montes et al., 2023). The robustness of RRM allows for missing records, but variance estimates at time points with sparse records may be more erratic (Nobre et al., 2003). Reliable estimates of genetic parameters for FE require sizable and quality feed intake data, which can be difficult to accumulate from research farms. However, Negussie et al. (2019) found that feed intake recording strategies that are sparse in terms of records per animal but with slightly more animals recorded per sire could still accurately estimate genetic parameters for DMI and RFI. In particular, DMI records that were collected once per week required between 8 and 46 daughters per sire to achieve breeding

value reliabilities ranging from 0.4 to 0.8, respectively. Among the 198 sires of the phenotyped calf population, the number of daughters per sire ranged from 1 to 66. Ultimately, this study provides a solid foundation for the validation of genetic parameters for daily MEI in future studies with larger datasets.

A critical consideration in genetic evaluations for feed efficiency is the timing of trait measurement. The RRM approach in this study highlights the dynamic nature of genetic influence on daily MEI across the first 2 mo of a calf's life, which further informs how to better select for an FE trait based on this measure of feed intake. Interestingly, genetic parameters were lowest during the transition period between the liquid and solid diet. This reduction may be attributable to inaccuracies in quantifying solid feed intake and challenges in distinguishing energy intake derived from milk versus solid feed during this period. Excluding the transition phase from genetic evaluations may therefore enhance the reliability of early-life feed efficiency traits. This approach also delineates 2 biologically distinct phases for evaluating calf feed intake, which could be used to develop an FE evaluation: (1) the pseudomonogastric phase, during which calves rely primarily on milk, and (2) the ruminant phase, following adaptation to solid feed. Recognizing these phases can support the development of more precise selection strategies for early-life FE in dairy cattle.

CONCLUSIONS

The results presented in this study provide insight into the variable genetic influence on feed intake of dairy calves during the rearing period. The genetic parameters estimated in this study changed over the rearing period, highlighting the need to consider how MEI, as well as the biological processes underpinning feeding and digestion, fluctuate as calves mature. The MEI is heritable and shows potential as a feed intake measurement for calf FE, however, further studies with larger datasets and additional herds are needed to confirm these findings.

NOTES

This study was part of the Resilient Dairy Genome Project (RDGP; <http://www.resilientdairy.ca/>) and the Net-Zero Dairy Genome Project (NDGP; <https://www.ndgp.ca>). We gratefully acknowledge all funding and support, as administered by Genome Canada (Ottawa, Canada), Genome Alberta (Calgary, Canada), Ontario Genomics (Toronto, Canada), Genome Quebec (Montreal, Canada), and Genome British Columbia (Vancouver, Canada). As per the research agreement, researchers maintain independence in conducting their studies, own their data, and report the outcomes regardless of the results. The deci-

sion to publish the findings rests solely with the researchers. Supplemental material for this article is available at <https://doi.org/10.5281/zenodo.18710387>. As per the research agreement, researchers maintain independence in conducting their studies, own their data, and report the outcomes regardless of the results. The decision to publish the findings rests solely with the researchers. This study was approved by the University of Guelph Animal Care Committee (AUP 4445 and AUP 5290). The authors have not stated any conflicts of interest.

Nonstandard abbreviations used: AIK = Akaike information criterion; CS = calf starter; FE = feed efficiency; GIT = gastrointestinal tract; MBW = metabolic BW; MEI = ME intake; MR = milk replacer; R1, R2 = research station nursery rooms; RFI = residual feed intake; RFID = radio frequency identification; RRM = random regression models.

REFERENCES

- Abuelo, A., F. Cullens, and J. L. Brester. 2021. Effect of preweaning disease on the reproductive performance and first-lactation milk production of heifers in a larger dairy herd. *J. Dairy Sci.* 104:7008–7017. <https://doi.org/10.3168/jds.2020-19791>.
- Baldwin, R. L. VI, K. R. McLeod, J. L. Klotz, and R. N. Heitmann. 2004. Rumen development, intestinal growth, and hepatic metabolism in the pre- and postweaning ruminant. *J. Dairy Sci.* 87:E55–E65. [https://doi.org/10.3168/jds.S0022-0302\(04\)70061-2](https://doi.org/10.3168/jds.S0022-0302(04)70061-2).
- Bazeley, K. 2003. Investigation of diarrhoea in the neonatal calf. In *Pract.* 25:152–159. <https://doi.org/10.1136/inpract.25.3.152>.
- Benetton, J. B., H. W. Neave, J. H. C. Costa, M. A. G. von Keyserlingk, and D. M. Weary. 2019. Automatic weaning based on individual solid feed intake: Effects on behavior and performance of dairy calves. *J. Dairy Sci.* 102:5475–5491. <https://doi.org/10.3168/jds.2018-15830>.
- Borderas, T. F., A. M. B. de Passillé, and J. Rushen. 2009. Feeding behavior of calves fed small or large amounts of milk. *J. Dairy Sci.* 92:2843–2852. <https://doi.org/10.3168/jds.2008-1886>.
- Conboy, M. H., C. B. Winder, M. C. Cantor, J. H. C. Costa, M. A. Steele, C. Medrano-Galarza, T. E. von Konigslow, A. Kerr, and D. L. Renaud. 2022. Association between feeding behaviours collected from an automatic milk feeder and neonatal calf diarrhea in group housed dairy calves: A case-control study. *Animals (Basel)* 12:170. <https://doi.org/10.3390/ani12020170>.
- Costa, J. H. C., M. C. Cantor, and H. W. Neave. 2021. Symposium review: Precision technologies for dairy calves and management applications. *J. Dairy Sci.* 104:1203–1219. <https://doi.org/10.3168/jds.2019-17885>.
- da Silva, C. S., J. M. Leão, C. F. A. Lage, S. G. Coelho, and M. M. Campos. 2023. Residual feed intake as an efficiency metric for pre-weaning dairy calves: What do we know? *Life (Basel)* 13:1727. <https://doi.org/10.3390/life13081727>.
- de Passillé, A. M., and J. Rushen. 2012. Adjusting the weaning age of calves fed by automated feeders according to individual intakes of solid feed. *J. Dairy Sci.* 95:5292–5298. <https://doi.org/10.3168/jds.2012-5521>.
- de Passillé, A. M., and J. Rushen. 2016. Using automated feeders to wean calves fed large amounts of milk according to their ability to eat solid feed. *J. Dairy Sci.* 99:3578–3583. <https://doi.org/10.3168/jds.2015-10259>.
- Diao, Q., R. Zhang, and T. Fu. 2019. Review of strategies to promote rumen development in calves. *Animals (Basel)* 9:490. <https://doi.org/10.3390/ani9080490>.
- Drackley, J. K. 2008. Calf nutrition from birth to breeding. *Vet. Clin. North Am. Food Anim. Pract.* 24:55–86. <https://doi.org/10.1016/j.cvfa.2008.01.001>.
- Ellingsen, K., C. M. Mejdell, N. Ottesen, S. Larsen, and A. M. Grøndahl. 2016. The effect of large milk meals on digestive physiology and behaviour in dairy calves. *Physiol. Behav.* 154:169–174. <https://doi.org/10.1016/j.physbeh.2015.11.025>.
- Fischer, T. M., A. R. Gilmour, and J. H. van der Werf. 2004. Computing approximate standard errors for genetic parameters derived from random regression models fitted by average information REML. *Genet. Sel. Evol.* 36:363–369. <https://doi.org/10.1186/1297-9686-36-363>.
- Flores, E. B., and J. van der Werf. 2015. Random regression test day models to estimate genetic parameters for milk yield and milk components in Philippine dairy buffaloes. *J. Anim. Breed. Genet.* 132:289–300. <https://doi.org/10.1111/jbg.12133>.
- Gelsinger, S. L., A. J. Heinrichs, and C. M. Jones. 2016. A meta-analysis of the effects of preweaned calf nutrition and growth on first-lactation performance. *J. Dairy Sci.* 99:6206–6214. <https://doi.org/10.3168/jds.2015-10744>.
- Ghaffari, M., H. Hammon, and C. Koch. 2025. Early rumen development in calves: biological processes and nutritional strategies—A mini review. *JDS Commun.* 6:427–431. <https://doi.org/10.3168/jdsc.2024-0702>.
- Gilmour, A. R., B. J. Gogel, B. R. Cullis, and R. Thompson. 2015. *ASReml User Guide - Release 4.2*. VSN International Ltd., Hemel Hempstead, UK.
- Gonzalez-Recio, O., J. E. Pryce, M. Haile-Mariam, and B. J. Hayes. 2014. Incorporating heifer feed efficiency in the Australian selection index using genomic selection. *J. Dairy Sci.* 97:3883–3893. <https://doi.org/10.3168/jds.2013-7515>.
- Graham, J. R., M. E. Montes, V. B. Pedrosa, J. Doucette, M. Taghipoor, A. C. Araujo, L. S. Gloria, J. P. Boerman, and L. F. Brito. 2024. Genetic parameters for calf feeding traits derived from automated milk feeding machines and number of bovine respiratory disease treatments in North American Holstein calves. *J. Dairy Sci.* 107:2175–2193. <https://doi.org/10.3168/jds.2023-23794>.
- Hammon, H. M., W. Liermann, D. Fieten, and C. Koch. 2020. Review: Importance of colostrum supply and milk feeding intensity on gastrointestinal and systemic development in calves. *Animal* 14:s133–s143. <https://doi.org/10.1017/S1751731119003148>.
- Hammon, H. M., G. Schiessler, A. Nussbaum, and J. W. Blum. 2002. Feed intake patterns, growth performance, and metabolic and endocrine traits in calves fed unlimited amounts of colostrum and milk by automate, starting in the neonatal period. *J. Dairy Sci.* 85:3352–3362. [https://doi.org/10.3168/jds.S0022-0302\(02\)74423-8](https://doi.org/10.3168/jds.S0022-0302(02)74423-8).
- Hawkins, A., K. Burdine, D. Amaral-Phillips, and J. H. C. Costa. 2019. An economic analysis of the costs associated with pre-weaning management strategies for dairy heifers. *Animals (Basel)* 9:471. <https://doi.org/10.3390/ani9070471>.
- Hayes, B. J., H. A. Lewin, and M. E. Goddard. 2013. The future of livestock breeding: Genomic selection for efficiency, reduced emissions intensity, and adaptation. *Trends Genet.* 29:206–214. <https://doi.org/10.1016/j.tig.2012.11.009>.
- Heinrichs, A. J., and B. S. Heinrichs. 2011. A prospective study of calf factors affecting first-lactation and lifetime milk production and age of cows when removed from the herd. *J. Dairy Sci.* 94:336–341. <https://doi.org/10.3168/jds.2010-3170>.
- Houlahan, K., F. S. Schenkel, D. Hailemariam, J. Lassen, M. Kargo, J. B. Cole, E. E. Connor, S. Wegmann, O. Junior, F. Miglior, A. Fleming, T. C. S. Chud, and C. F. Baes. 2021. Effects of incorporating dry matter intake and residual feed intake into a selection index for dairy cattle using deterministic modeling. *Animals (Basel)* 11:1157. <https://doi.org/10.3390/ani11041157>.
- Houlahan, K., F. S. Schenkel, F. Miglior, J. Jamrozik, R. B. Stephansen, O. González-Recio, N. Charfeddine, D. Segelke, A. M. Butty, P. Stratz, M. J. VandeHaar, R. J. Tempelman, K. Weigel, H. White, F. Peñagaricano, J. E. Koltes, J. E. P. Santos, R. L. Baldwin, and C. F. Baes. 2024. Estimation of genetic parameters for feed efficiency

- traits using random regression models in dairy cattle. *J. Dairy Sci.* 107:1523–1534. <https://doi.org/10.3168/jds.2022-23124>.
- Hyland, E. 2022. Evaluating calf health recording and incidence of respiratory disease and diarrhea on Ontario dairy farms using producer recorded data. MS thesis. Department of Animal Biosciences, University of Guelph, Guelph, ON, Canada. <https://hdl.handle.net/10214/26675>.
- Jahnel, R. E., I. Blunk, D. Wittenburg, and N. Reinsch. 2023. Relationship between milk urea content and important milk traits in Holstein cattle. *Animal* 17:100767. <https://doi.org/10.1016/j.animal.2023.100767>.
- Jamrozik, J., and L. R. Schaeffer. 1997. Estimates of genetic parameters for a test day model with random regressions for yield traits of first lactation Holsteins. *J. Dairy Sci.* 80:762–770. [https://doi.org/10.3168/jds.S0022-0302\(97\)75996-4](https://doi.org/10.3168/jds.S0022-0302(97)75996-4).
- Jensen, M. B. 2007. Age at introduction to the group affects dairy calves' use of a computer-controlled milk feeder. *Appl. Anim. Behav. Sci.* 107:22–31. <https://doi.org/10.1016/j.applanim.2006.09.017>.
- Karacaören, B., F. Jaffrézic, and H. N. Kadarmideen. 2006. Genetic parameters for functional traits in dairy cattle from daily random regression models. *J. Dairy Sci.* 89:791–798. [https://doi.org/10.3168/jds.S0022-0302\(06\)72141-5](https://doi.org/10.3168/jds.S0022-0302(06)72141-5).
- Kennedy, B. W., J. H. van der Werf, and T. H. Meuwissen. 1993. Genetic and statistical properties of residual feed intake. *J. Anim. Sci.* 71:3239–3250. <https://doi.org/10.2527/1993.71123239x>.
- Khan, M. A., A. Bach, D. M. Weary, and M. A. G. von Keyserlingk. 2016. Invited review: Transitioning from milk to solid feed in dairy heifers. *J. Dairy Sci.* 99:885–902. <https://doi.org/10.3168/jds.2015-9975>.
- Khan, M. A., H. J. Lee, W. S. Lee, H. S. Kim, S. B. Kim, K. S. Ki, J. K. Ha, H. G. Lee, and Y. J. Choi. 2007. Pre- and postweaning performance of Holstein female calves fed milk through step-down and conventional methods. *J. Dairy Sci.* 90:876–885. [https://doi.org/10.3168/jds.S0022-0302\(07\)71571-0](https://doi.org/10.3168/jds.S0022-0302(07)71571-0).
- Khan, M. A., D. M. Weary, and M. A. G. Von Keyserlingk. 2011. Hay intake improves performance and rumen development of calves fed higher quantities of milk. *J. Dairy Sci.* 94:3547–3553. <https://doi.org/10.3168/jds.2010-3871>.
- Khanal, P., J. Johnson, G. Gouveia, A. T. H. Utsunomiya, P. Ross, and N. Deeb. 2024. Genomic evaluation of residual feed intake in US Holstein cows: Insights into lifetime feed efficiency. *Front. Genet.* 15:1462306. <https://doi.org/10.3389/fgene.2024.1462306>.
- Khanal, P., K. L. Parker Gaddis, M. J. Vandehaar, K. A. Weigel, H. M. White, F. Peñagaricano, J. E. Koltes, J. E. P. Santos, R. L. Baldwin, J. F. Burchard, J. W. Dürr, and R. J. Tempelman. 2022. Multiple-trait random regression modeling of feed efficiency in US Holsteins. *J. Dairy Sci.* 105:5954–5971. <https://doi.org/10.3168/jds.2021-21739>.
- Khorshidi, R., M. D. MacNeil, D. P. Hays, M. K. Abo-Ismael, J. J. Crowley, E. C. Akanno, Z. Wang, and G. Plastow. 2020. Estimation of genetic parameters and trends for growth traits in Hays Converter cattle using multiple-trait and random regression models. *Livest. Sci.* 241:104245. <https://doi.org/10.1016/j.livsci.2020.104245>.
- Kirkpatrick, M., D. Lofsvold, and M. Bulmer. 1990. Analysis of the inheritance, selection and evolution of growth trajectories. *Genetics* 124:979–993. <https://doi.org/10.1093/genetics/124.4.979>.
- Koch, R. M., L. A. Swiger, D. Chambers, and K. E. Gregory. 1963. Efficiency of feed use in beef cattle. *J. Anim. Sci.* 22:486–494. <https://doi.org/10.2527/jas1963.222486x>.
- Krachun, C., J. Rushen, and A. M. de Passillé. 2010. Play behaviour in dairy calves is reduced by weaning and by a low energy intake. *Appl. Anim. Behav. Sci.* 122:71–76. <https://doi.org/10.1016/j.applanim.2009.12.002>.
- Leal, L. N., J. B. Daniel, J. Doelman, B. R. Keppler, M. A. Steele, and J. Martín-Tereso. 2025. Prewaning nutrient supply improves lactation productivity and reduced the risk of culling in Holstein cows. *J. Dairy Sci.* 108:5875–5888. <https://doi.org/10.3168/jds.2024-26102>.
- Li, B., W. F. Fikse, P. Løvendahl, J. Lassen, M. H. Lidauer, P. Mäntysaari, and B. Berglund. 2018. Genetic heterogeneity of feed intake, energy-corrected milk, and body weight across lactation in primiparous Holstein, Nordic Red, and Jersey cows. *J. Dairy Sci.* 101:10011–10021. <https://doi.org/10.3168/jds.2018-14611>.
- Meale, S. J., F. Chaucheyras-Durand, H. Berends, L. L. Guan, and M. A. Steele. 2017. From pre- to postweaning: Transformation of the young calf's gastrointestinal tract. *J. Dairy Sci.* 100:5984–5995. <https://doi.org/10.3168/jds.2016-12474>.
- Medrano-Galarza, C., S. J. LeBlanc, T. J. DeVries, A. Jones-Bitton, J. Rushen, A. M. de Pasille, M. I. Endres, and D. B. Haley. 2018. Effect of age on introduction to an automated milk feeder on calf learning and performance and labor requirements. *J. Dairy Sci.* 101:9371–9384. <https://doi.org/10.3168/jds.2018-14390>.
- Medrano-Galarza, C., S. J. LeBlanc, T. J. DeVries, A. Jones-Bitton, J. Rushen, A. M. de Pasille, and D. B. Haley. 2017. A survey of dairy calf management practices among farms using manual and automated milk feeding systems in Canada. *J. Dairy Sci.* 100:6872–6884. <https://doi.org/10.3168/jds.2016-12273>.
- Miller-Cushon, E. K., R. Bergeron, K. E. Leslie, and T. J. DeVries. 2013a. Effect of milk feeding level on development of feeding behavior in dairy calves. *J. Dairy Sci.* 96:551–564. <https://doi.org/10.3168/jds.2012-5937>.
- Miller-Cushon, E. K., R. Bergeron, K. E. Leslie, G. J. Mason, and T. J. DeVries. 2013b. Effect of early exposure to different feed presentations on feed sorting of dairy calves. *J. Dairy Sci.* 96:4624–4633. <https://doi.org/10.3168/jds.2013-6670>.
- Miller-Cushon, E. K., and T. J. DeVries. 2015. Invited review: Development and expression of dairy calf feeding behaviour. *Can. J. Anim. Sci.* 95:341–350. <https://doi.org/10.4141/cjas-2014-163>.
- Montes, M. E., J. Doucette, L. F. Brito, and J. P. Boerman. 2023. Environmental and biological factors that influence behavior of Holstein calves in automated milk feeding systems. *JDS Commun.* 4:379–384. <https://doi.org/10.3168/jdsc.2023-0374>.
- Neave, H. W., J. H. C. Costa, J. B. Benetton, D. M. Weary, and M. A. G. von Keyserlingk. 2019. Individual characteristics in early life relate to variability in weaning age, feeding behavior, and weight gain of dairy calves automatically weaned based on solid feed intake. *J. Dairy Sci.* 102:10250–10265. <https://doi.org/10.3168/jds.2019-16438>.
- Neave, H. W., J. H. C. Costa, D. M. Weary, and M. A. G. von Keyserlingk. 2018. Personality is associated with feeding behavior and performance in dairy calves. *J. Dairy Sci.* 101:7437–7449. <https://doi.org/10.3168/jds.2017-14248>.
- Negussie, E., T. Mehtio, P. Mäntysaari, P. Løvendahl, E. A. Mäntysaari, and M. H. Lidauer. 2019. Reliability of breeding values for feed intake and feed efficiency traits in dairy cattle: When dry matter intake recordings are sparse under different scenarios. *J. Dairy Sci.* 102:7248–7262. <https://doi.org/10.3168/jds.2018-16020>.
- NFACC (National Farm Animal Council). 2023. Code of practice for the care and handling of dairy cattle. National Farm Animal Care Council, Lacombe, AB, Canada. Accessed Jan. 27, 2025. <https://www.nfacc.ca/codes-of-practice/dairy-cattle>.
- Nobre, P. R. C., I. Misztal, S. Tsuruta, J. K. Bertrand, L. O. C. Silva, and P. S. Lopes. 2003. Analyses of growth curves of Nellore cattle by multiple-trait and random regression models. *J. Anim. Sci.* 81:918–926. <https://doi.org/10.2527/2003.814918x>.
- Oliveira, H. R., L. F. Brito, D. A. L. Lourenco, F. F. Silva, J. Jamrozik, L. R. Schaeffer, and F. S. Schenkel. 2019. Invited review: Advances and applications of random regression models: From quantitative genetics to genomics. *J. Dairy Sci.* 102:7664–7683. <https://doi.org/10.3168/jds.2019-16265>.
- Ollivett, T. L., D. V. Nydam, T. C. Linden, D. D. Bowman, and M. E. Van Amburgh. 2012. Effect of nutritional plane on health and performance in dairy calves after experimental infection with *Cryptosporidium parvum*. *J. Am. Vet. Med. Assoc.* 241:1514–1520. <https://doi.org/10.2460/javma.241.11.1514>.
- Paneru, U., N. Moghaddar, and J. van der Werf. 2024. Comparison between multiple-trait and random regression models for genetic evaluation of weight traits in Australian meat sheep. *J. Anim. Sci.* 102:skae038. <https://doi.org/10.1093/jas/skae038>.
- Parsons, S. D., M. A. Steele, K. E. Leslie, D. L. Renaud, C. N. Reedman, C. B. Winder, and T. J. DeVries. 2022. Effect of a milk byproduct-based calf starter feed on dairy calf nutrient consumption, rumen development, and performance when fed different milk levels. *J. Dairy Sci.* 105:281–300. <https://doi.org/10.3168/jds.2021-21018>.

- Pereira, R. J., A. B. Bignardi, L. El Faro, R. S. Verneque, A. E. Vercesi Filho, and L. G. Albuquerque. 2013. Random regression models using Legendre polynomials or linear splines for test-day milk yield of dairy Gyr (*Bos indicus*) cattle. *J. Dairy Sci.* 96:565–574. <https://doi.org/10.3168/jds.2011-5051>.
- Pryce, J. E., O. Gonzalez-Recio, G. Nieuwhof, W. J. Wales, M. P. Coffey, B. J. Hayes, and M. E. Goddard. 2015. Hot topic; Definition and implementation of a breeding value for feed efficiency in dairy cows. *J. Dairy Sci.* 98:7430–7350. <https://doi.org/10.3168/jds.2015-9621>.
- Quigley, J. D., L. Deikun, T. M. Hill, F. X. Suarez-Mena, T. S. Dennis, and W. Hu. 2019. Effects of colostrum and milk replacer feeding rates on intake, growth, and digestibility in calves. *J. Dairy Sci.* 102:11016–11025. <https://doi.org/10.3168/jds.2019-16682>.
- R Core Team. 2022. R: A language and environment for statistical computing. R Foundation for Statistical Computing. <https://www.r-project.org/>.
- Rosenberger, K., J. H. C. Costa, H. W. Neave, M. A. G. von Keyserlingk, and D. M. Weary. 2017. The effect of milk allowance on behavior and weight gains in dairy calves. *J. Dairy Sci.* 100:504–512. <https://doi.org/10.3168/jds.2016-11195>.
- Roth, B. A., N. M. Keil, L. Gyax, and E. Hillmann. 2009. Influence of weaning method on health status and rumen development in dairy calves. *J. Dairy Sci.* 92:645–656. <https://doi.org/10.3168/jds.2008-1153>.
- Schaeffer, L. R. 2004. Application of random regression models in animal breeding. *Livest. Prod. Sci.* 86:35–45. [https://doi.org/10.1016/S0301-6226\(03\)00151-9](https://doi.org/10.1016/S0301-6226(03)00151-9).
- Seymour, D. J., A. Canovas, C. F. Baes, T. C. S. Chud, V. R. Osborne, J. P. Cant, L. F. Brito, B. Gredler-Grandl, R. Finocchiaro, R. F. Veerkamp, Y. de Haas, and F. Miglior. 2019. Invited review: Determination of large-scale individual dry matter intake phenotypes in dairy cattle. *J. Dairy Sci.* 102:7655–7663. <https://doi.org/10.3168/jds.2019-16454>.
- Silva, A. L., T. J. Devries, L. O. Tedeschi, and M. I. Marcondes. 2019. Development of equations, based on milk intake, to predict starter feed intake of preweaned dairy calves. *Animal* 13:83–89. <https://doi.org/10.1017/S1751731118000666>.
- Soberon, F., E. Raffenato, R. W. Everett, and M. E. Van Amburgh. 2012. Prewaning milk replacer intake and effects on long-term productivity of dairy calves. *J. Dairy Sci.* 95:783–793. <https://doi.org/10.3168/jds.2011-4391>.
- Strucken, E. M., Y. C. S. M. Laurenson, and G. A. Brockmann. 2015. Go with the flow-biology and genetics of the lactation cycle. *Front. Genet.* 6. <https://doi.org/10.3389/fgene.2015.00118>.
- Sweeney, B. C., J. Rushen, D. M. Weary, and A. M. de Passillé. 2010. Duration of weaning, starter intake, and weight gain of dairy calves fed large amounts of milk. *J. Dairy Sci.* 93:148–152. <https://doi.org/10.3168/jds.2009-2427>.
- Togashi, K., C. Y. Lin, Y. Atagi, K. Hagiya, J. Sato, and T. Nakanishi. 2008. Genetic characteristics of Japanese Holstein cows based on multiple-lactation random regression test-day animal models. *Livest. Sci.* 114:194–201. <https://doi.org/10.1016/j.livsci.2007.04.023>.
- Van Amburgh, M. E., F. Soberon, M. J. Meyer, and R. A. Molano. 2019. Symposium review: Integration of postweaning nutrient requirements and supply with composition of growth and mammary development in modern dairy heifers. *J. Dairy Sci.* 102:3692–3705. <https://doi.org/10.3168/jds.2018-15270>.
- Van De Stroet, D. L., J. A. Calderón Díaz, K. J. Stalder, A. J. Heinrichs, and C. D. Dechow. 2016. Association of calf growth traits with production characteristics in dairy cattle. *J. Dairy Sci.* 99:8347–8355. <https://doi.org/10.3168/jds.2015-10738>.
- van Staaveren, N., E. Hyland, K. Houlahan, C. Lynch, F. Miglior, D. F. Kelton, F. S. Schenkel, and C. F. Baes. 2023. Recording of calf diseases for potential use in breeding programs: A case study on calf respiratory illness and diarrhea. *Can. J. Anim. Sci.* 103:192–203. <https://doi.org/10.1139/cjas-2022-0112>.
- Welk, A., H. W. Neave, H. B. Spitzer, M. A. G. von Keyserlingk, and D. M. Weary. 2022. Effects of intake-based weaning and forage type on feeding behavior and growth of dairy calves fed by automated feeders. *J. Dairy Sci.* 105:9119–9136. <https://doi.org/10.3168/jds.2021-21468>.
- Wolc, A., E. Barczak, J. Wójtowski, P. Ślósarz, and T. Szwaczkowski. 2011. Genetic parameters of body weight in sheep estimated via random regression and multi-trait animal models. *Small Rumin. Res.* 100:15–18. <https://doi.org/10.1016/j.smallrumres.2011.05.009>.
- Yonekura, S., K. Kitade, G. Furukawa, K. Takahashi, N. Katsumata, K. Katoh, and Y. Obara. 2002. Effects of aging and weaning on mRNA expression of leptin and CCK receptors in the calf rumen and abomasum. *Domest. Anim. Endocrinol.* 22:25–35. [https://doi.org/10.1016/S0739-7240\(01\)00114-X](https://doi.org/10.1016/S0739-7240(01)00114-X).

ORCID

- A. G. R. Phillips, <https://orcid.org/0009-0003-8570-4874>
 B. O. Makanjuola, <https://orcid.org/0000-0002-3485-0201>
 F. Miglior, <https://orcid.org/0000-0003-2345-8842>
 F. S. Schenkel, <https://orcid.org/0000-0001-8700-0633>
 M. A. Steele, <https://orcid.org/0000-0001-6941-6205>
 C. F. Baes, <https://orcid.org/0000-0001-6614-8890>
 R. E. Jahnel, <https://orcid.org/0000-0002-4638-2046>